Thermodynamic Modeling: Ethylenediaminetetraacetic acid tetrasodium salt (Na4EDTA)

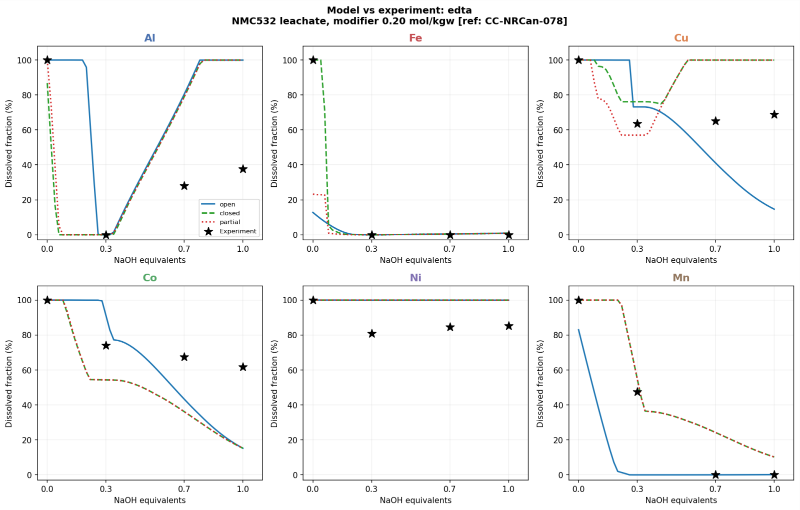
NMC532 Leachate, Modifier 0.20 mol/kgw

MDPSA Pipeline

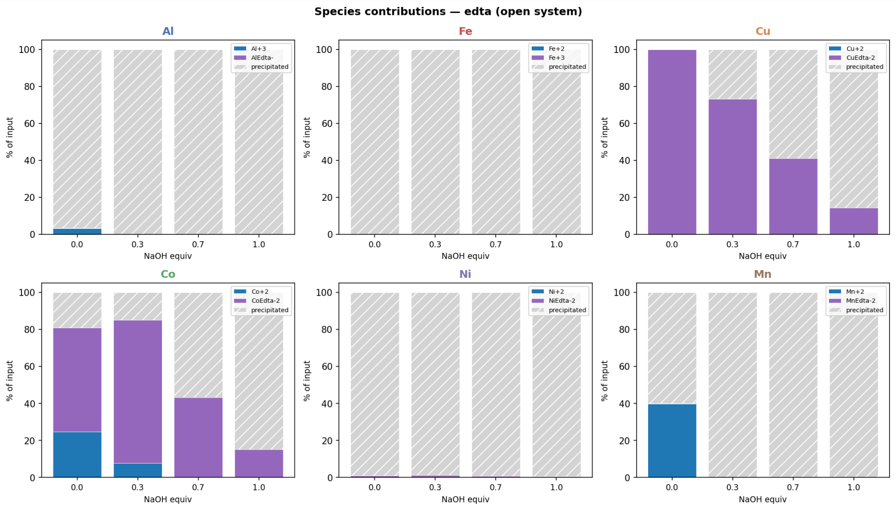
2026-07-22

NRCan CC-NRCan-078

Model vs. Experiment



Species Contributions (Open System)



Selectivity Scorecard

Metal	Role	@ 0.3 eq	@ 1.0 eq	RMSE (pp)	Verdict
Cu	Target	73%	15%	30.0	Partial
Co	Target	91%	15%	27.6	Partial
Ni	Target	100%	100%	14.3	Protected
Mn	Target	0%	0%	25.2	Lost
Al	Impurity	0%	100%	40.7	Partial
Fe	Impurity	0%	1%	43.6	Removed

RMSE: root-mean-square error in percentage points vs. experiment (CC-NRCan-078).
Protected/Removed: model matches experiment at all comparison equiv points.

Thermodynamic Parameters: Ethylenediaminetetraacetic acid tetrasodium salt (Na4EDTA)

Species	log K
CuEdta-2	20.50
CoEdta-2	16.31
NiEdta-2	18.62
MnEdta-2	13.87
AlEdta-	16.30

- Database: sit.dat (SIT activity model)
- Modifier dose: 0.20 mol/kgw
- Temperature: 25°C, I = 0 (NIST reference)
- Equilibrium phases: Gibbsite, Ferrihydrite(am), $Cu(OH)_2(s)$, $Co(OH)_2(s)$, $Ni(OH)_2(s)$, MnOOH
- Fe complexation suppressed (kinetic)
- Scenarios: open / closed / partial CO_2

Source: NIST Standard Reference Database 46 Version 8.0. Values adjusted for ionic-strength effects at leachate conditions.

MDPSA: Multi-Agent Dissolution–Precipitation Simulation

1. **ScoutAgent** : queries Claude (LLM) for NIST formation constants; writes `.phr` (PHREEQC reactions) and `.yaml` (species list).
2. **SimRunnerAgent** : executes PHREEQC for three CO_2 scenarios (open, closed, partial); parses output CSVs.
3. **AnalystAgent** : computes dissolved-metal fractions, RMSE vs. experiment, selectivity scorecard, and species-contribution figures.
4. **BOAgent** : Bayesian optimization (Gaussian process) over $\log K$ space; suggests next candidate modifier.
5. **BeamerAgent** : assembles and compiles this XeLaTeX presentation automatically.

Key design choices

- LangGraph `StateGraph` orchestration
- SIT activity model (sit.dat)
- NaOH titration: $0 \rightarrow 1.0$ equiv in 60 steps
- Modifier dose fixed at 0.20 mol/kgw
- Comparison at 0.3 and 1.0 equiv NaOH
- Target metals: Cu, Co, Ni; impurities: Al, Fe, Mn

Output

- Model-vs-experiment figure
- Species-contribution breakdown
- Selectivity scorecard (RMSE, verdict)
- Auto-compiled PDF (this file)

Pipeline: LangGraph StateGraph; LLM: Anthropic Claude (claude-sonnet-4-5). PHREEQC v3, TeX Live 2026.